

RENANCE

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PEE 514 — Past Questions & Materials
by Renance

Course: PEE 514
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 500

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Paper 1 of 1 — PEE 514 - Petroleum Engineering Analysis

Level 500 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
FIRST SEMESTER EXAMINATION, 2018/2019 SESSION
PNG 514: PETROLEUM SYSTEM ANALYSIS

TIME ALLOWED: 3 HRS DATE: 06-06-2019 UNIT: 3

INSTRUCTION: ANSWER TWO QUESTIONS FROM EACH SECTION

SECTION A

QUESTION 1

Approximate the integral $\int_0^3 (x+1)^2 \ln 3x \, dx$ with four intervals by using

(a) Trapezoidal rule to calculate the absolute relative error
(b) Simpson's 1/3 rule to calculate the absolute relative error

QUESTION 2

Given the pressure-volume relationship as

$$PV^k = C$$

V	2	4	6	8	10	12
P	101.2	49.2	37.6	28.4	19.2	10.1

Find P when V = 100

QUESTION 3

Given the figure below with five block of equal size,

Additional information: $k = 62.5 \text{ W/m-K}$, density = 8120 kg/m^3 , $C = 520 \text{ J/kg-K}$, Initial temperature = 105°C

The equation that governs the heat conduction is given as

$$\frac{\partial^2 T}{\partial x^2} = \frac{\rho C}{k} \frac{\partial T}{\partial t}$$

The temperature distribution in the slab was monitored from $t = 0$ to 12 seconds with time step of 4 seconds. Find the temperature distribution at the first time step respectively using the explicit formulation.

$$\int_a^b f(x) dx = \frac{b-a}{3n} \left[f(x_0) + 4 \sum_{i=1}^{n-1} f(x_i) + 2 \sum_{i=2}^{n-2} f(x_i) + f(x_n) \right]$$

$$\int_a^b f(x) dx = \frac{b-a}{8} [f(x_0) + 3(f(x_2) + f(x_4) + f(x_6) + \dots + f(x_n))]$$

$$\int_a^b f(x) dx = \frac{b-a}{2n} \left[f(a) + 2 \sum_{i=1}^{n-1} f(a+ih) + f(b) \right]$$

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SECTION B

QUESTION 4

- (a) State five key areas where mathematical techniques have helped in the development of the world today.
- (b) Derive the diffusivity equation for a slightly compressible fluid for a single phase radial system from the law of conservation of mass, neglecting the diffusion and reaction terms. Also assume a homogenous and isotropic medium with no injection of fluid into the control volume.

QUESTION 5

Given the following saturation and reservoir water cut values in the table below

S_{WC}	T_{WC}	T_{WC}	T_{WC}	T_{WC}
0.29	0.35	0.42	0.5	
0.0018	0.0149	0.0394	0.249	

- (a) Apply Lagrange interpolation method to determine the water cut f_w at 0.58

- (b) Draw the various plot of the Lagrange interpolant $l_0(x)$, $l_1(x)$, $l_2(x)$ and a combination of the three showing the location of the roots.

- (c) Use the water cut and saturation data above, apply Newton's divided-difference approximation to estimate the approximate function of the water cut at 0.58 saturation value for a second degree polynomial fitting

$$(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_{i-1}) \dots (x_1 - x_N)$$

QUESTION 6

The following equation describes how the temperature gradient varies with the length of the pipeline. The temperature gradient has a root at zero (0) and the stop of the points (1.2, 2.4) & (5, 32.8) along the pipeline.

$$\frac{dT}{dL} = f(L) = L^3 - 8.6L^2 + 4.925 = 0$$

Conduct several iterations to estimate the root of the above equation until the absolute relative error is less than 35%.

Hint:

$$g(x) = f(x_0) + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] \\ + (x - x_0)(x - x_1)(x - x_2)f[x_0, x_1, x_2, x_3]$$

$$f[x_0, x_1, \dots, x_i] = \frac{f[x_0, \dots, x_{i-1}] - f[x_1, x_i]}{x_0 - x_i}.$$

$$g(x) = \sum_{l=0}^N f_l I_l(x)$$

$$\int_0^b f(x) dx = \frac{b-a}{2n} \left[f(0) + 2 \sum_{i=1}^{n-1} f(a+ih) + f(b) \right]$$

$$I_1(x) = \frac{(x-x_0)(x-x_1)(x-x_2)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_N)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_{i-1})(x_1-x_{i+1})\dots(x_1-x_N)}$$

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (21)^2 \cdot 10 = 9240 \text{ cm}^3$$

$$1) \int_0^3 (x+1)^2 \ln 3x \, dx \quad \text{with } n=4$$

Analytical method

$$\int u \, dv = uv - \int v \, du$$

$$\text{let } u = \ln 3x \quad dv = (x+1)^2 \, dx$$

$$\text{From } u = \ln 3x$$

$$\text{let } w = 3x \quad ; \quad dw/dx = 3$$

$$u = \ln w$$

$$du/dw = 1/w$$

$$du/dx = du/dw \times dw/dx = 1/w \times 3 = 1/3x \times 3 = 1/x \quad \left\{ du = dx/x \right.$$

$$v = \int (x+1)^2 \, dx = \frac{1}{\frac{d}{dx}(x+1)} \cdot \frac{(x+1)^{2+1}}{2+1} = \frac{1}{1} \cdot \frac{(x+1)^3}{3} = \frac{(x+1)^3}{3}$$

$$\begin{aligned} \int u \, dv &= \ln 3x \cdot \frac{(x+1)^3}{3} - \int \frac{(x+1)^3}{3} \cdot \frac{1}{x} \, dx \quad \left\{ \begin{aligned} (x+1)^3 &= (x+1)^2(x+1) \\ &= (x^2+2x+1)(x+1) \\ &= x^3+x^2+2x^2+2x+x+1 \\ &= x^3+3x^2+3x+1 \end{aligned} \right. \\ &= \ln 3x \cdot \frac{(x+1)^3}{3} - \frac{1}{3} \int \frac{x^3+3x^2+3x+1}{x} \, dx \\ &= \ln 3x \cdot \frac{(x+1)^3}{3} - \frac{1}{3} \int \left(x^2+3x+3+\frac{1}{x} \right) \, dx \\ &= \ln 3x \cdot \frac{(x+1)^3}{3} - \left[\frac{x^3}{9} + \frac{3x^2}{6} + \frac{3x}{3} + \frac{\ln x}{3} \right] \end{aligned}$$

Applying the limit

$$\begin{aligned} &\left[\ln 3x \cdot \frac{(x+1)^3}{3} \right]_0^3 - \left[\frac{x^3}{9} + \frac{3x^2}{6} + \frac{3x}{3} + \frac{\ln x}{3} \right]_0^3 \\ &= \left[\ln(3 \times 3) \cdot \frac{4^3}{3} - 0 \right] - \left[\frac{27}{9} + \frac{9}{2} + 3 + \frac{\ln(3)}{3} - 0 \right] \\ &= \underline{46.87412} - 10.86620 \\ &= \underline{36.00792} \end{aligned}$$

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Using Trapezoidal Rule $\int_a^b f(x) = \frac{b-a}{2n} \left(f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4) \right)$

where $h = \frac{b-a}{n}$

from $\int_0^3 (x+1)^2 \ln(3x) dx$ $h = \frac{3-0}{4} = 0.75$

$x_0 = 0, x_1 = 0.75, x_2 = 1.5, x_3 = 2.25, x_4 = 3$

$f(x_0) = f(x_0) = (0+1)^2 \ln(3 \times 0) = 0$;

$f(x_1) = f(x_1) = (0.75+1)^2 \ln(3 \times 0.75) = 2.483474$

$f(x_2) = f(x_2) = (1.5+1)^2 \ln(3 \times 1.5) = 9.40048$

$f(x_3) = f(x_3) = (2.25+1)^2 \ln(3 \times 2.25) = 20.16954$

$f(x_4) = f(x_4) = (3+1)^2 \ln(3 \times 3) = 35.155593$

$$\int_a^b f(x) = \frac{3-0}{2(4)} \left(0 + 2(2.483474) + 2(9.40048) + 2(20.16954) + 35.155593 \right)$$

$$= \frac{3}{8} \left[99.262581 \right] = \underline{\underline{37.2234}}$$

Now the absolute relative error = $\frac{\text{true value} - \text{approx value}}{\text{true value}} \times 100$

$$= \left| \frac{36.00792 - 37.2234}{36.00792} \right| \times 100$$

$$= \underline{\underline{3.376\%}}$$

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1 b) using Simpson's $1/3$ rule

$$\int_a^b f(x) dx = \frac{b-a}{3n} \left[f(x_0) + 4(f(x_1) + f(x_3)) + 2(f(x_2) + f(x_4)) \right]$$

$$h = \frac{b-a}{n} = \frac{3-0}{4} = 0.75$$

$$f(x_0) = 0 ; f(x_1) = 2.483474 \quad f(x_2) = 9.40048$$

$$f(x_3) = 20.16954 \quad f(x_4) = 35.155593$$

$$\int_a^b f(x) dx = \frac{3-0}{(4)} \left[0 + 4(2.483474 + 20.16954) + 2(9.40048 + 35.155593) \right]$$

$$\int_a^b f(x) dx = \underline{\underline{36.1421523}}$$

Now the absolute relative error

$$= \frac{\text{true value} - \text{approx value}}{\text{true value}} \times 100$$

$$= \left| \frac{36.00792 - 36.14215}{36.00792} \right| \times 100$$

$$= \left| \frac{36.00792 - 36.1421523}{36.00792} \right| \times 100$$

$$= 0.37279\%$$

$$= \underline{\underline{0.37279\%}}$$

QUESTION 2

$$PV^k = C$$

Soln

$P = CV^{-k}$ Taking log of both side {NB: if the eqn involves exponents take ln of both sides}

$$\log P = \log C + (-k) \log V \quad \left\{ \begin{array}{l} \text{let } Y = \log P \\ \text{let } a = \log C \\ \text{let } X = \log V \\ b = -k \end{array} \right.$$

$Y = a + bX \Rightarrow$ The equation to the solution is

$$\sum Y = na + b \sum X \quad \dots \times$$

$$\sum XY = a \sum X + b \sum X^2 \quad \dots \times$$

	V	$Y = \log P$	$X = \log V$	XY	X^2
1.2	54.3	1.73675	1.73480	3.0097	3.0095
9.2	61.8	1.69197	1.7910	3.0303	3.20768
7.6	72.4	1.57519	1.8597	2.9294	3.45848
8.4	88.7	1.4533	1.9479	2.8309	3.7943
9.2	118.6	1.2833	2.07408	2.66167	4.3018
0.1	194	1.0043	2.2878	2.2976	5.2340
		$\sum Y = 8.79481$	$\sum X = 11.69528$	$\sum XY = 16.84957$	$\sum X^2 = 23.00576$

Where $n=6$ substitute the value of $\sum Y, \sum X, \sum XY, \sum X^2$ into the above eqn

$$6a + 11.69528b = 8.79481 \quad \dots \textcircled{1}$$

$$11.69528a + 23.00576b = 16.84957 \quad \dots \textcircled{2}$$

$$\begin{bmatrix} n & \sum X \\ \sum X & \sum X^2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sum Y \\ \sum XY \end{bmatrix}$$

Solving in matrix form or simultaneously

{ For this purpose Natural display calculator was used to solve the simultaneous eqn. please ensure you solve it yourself }

Y/ban 9.5

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2 CONT'D

$$a = 4.19993 \quad b = -1.4027$$

Now $\log C = a$

$$\log C = 4.19993$$

$$C = \underline{\underline{15823.41}}$$

Also $b = -K$

$$(-K) = -1.4027$$

Hence $P = 15823.41 V^{-1.4027}$

When $V = 100$

$$P = 15823.41 (100)^{-1.4027}$$

$$P = \underline{\underline{24.77 \text{ psi}}}$$

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UET-3

Soln

$$\frac{\partial^2 T}{\partial x^2} = \frac{\rho C}{K} \frac{\partial T}{\partial t} \quad \begin{cases} \Delta t = 4 \text{ sec} \\ \Delta x = 2.75 \frac{\text{cm}}{100} = \underline{\underline{0.0275 \text{ m}}} \end{cases}$$

Explicit Formulation implies

$$\left(\frac{\partial^2 T}{\partial x^2} \right)_i^n = \frac{\rho C}{K} \left(\frac{\partial T}{\partial t} \right)_i^n$$

$$\frac{T_{i-1}^n - 2T_i^n + T_{i+1}^n}{(\Delta x)^2} = \frac{\rho C}{K} \frac{T_i^{n+1} - T_i^n}{\Delta t}$$

Now

$$\frac{K \Delta t}{\rho C (\Delta x)^2} \times [T_{i-1}^n - 2T_i^n + T_{i+1}^n] = T_i^{n+1} - T_i^n \quad \dots (i)$$

$$\text{let } \beta = \frac{K \Delta t}{\rho C (\Delta x)^2} = \frac{62.5 \times 4}{8120 \times 520 \times (0.0275)^2} = 0.07829$$

$$\text{Number of time step} = \frac{t_{\text{final}} - t_{\text{initial}}}{\Delta t} = \frac{(12 - 0) \text{ sec}}{4 \text{ sec}} = 3 //$$

The question would be solved in 4 nodal time step

$$t=0, t=4 \text{ sec } t=8 \text{ sec } t=12 \text{ sec}$$

From eqn (i)

$$\beta [T_{i-1}^n - 2T_i^n + T_{i+1}^n] = T_i^{n+1} - T_i^n$$

$$\beta T_{i-1}^n - 2\beta T_i^n + \beta T_{i+1}^n = T_i^{n+1} - T_i^n$$

$$\beta T_{i-1}^n + T_i^n - 2\beta T_i^n + \beta T_{i+1}^n = T_i^{n+1}$$

$$\beta T_{i-1}^n + (1 - 2\beta) T_i^n + \beta T_{i+1}^n = T_i^{n+1} \quad \{ \beta = 0.07829 \}$$

(7)

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Boundary Conditions

$$T_0^n = 235^\circ\text{C} \text{ for all } n = 0, 1, 2, 3, 4$$

$$T_5^n = 138^\circ\text{C} \text{ for all } n = 0, 1, 2, 3, 4$$

$$T_{(1-4)}^0 = 105^\circ\text{C} \text{ i.e. } T_1^0 = 105^\circ\text{C}, T_2^0 = 105^\circ\text{C}, T_3^0 = 105^\circ\text{C}, T_4^0 = 105^\circ\text{C}$$

We are to find temp distribution at the first time step, we will stop at $t = 4 \text{ sec}$

Now at $t = 0 \text{ sec}$, i.e., the nodal temperatures are

$$T_0^0 = 235^\circ\text{C}, T_1^0 = 105^\circ\text{C}, T_2^0 = 105^\circ\text{C}, T_3^0 = 105^\circ\text{C}, T_4^0 = 105^\circ\text{C}, T_5^0 = 138^\circ\text{C}$$

To get the interior node set $i = 1, 2, 3, 4$ & $n = 0$

$$\text{for } i = 1, n = 0 \quad \Delta \tau = 0.07829, T_0^0 = 235^\circ\text{C}, T_1^0 = 105^\circ\text{C}, T_2^0 = 105^\circ\text{C}$$

$$T_i^{n+1} = \beta T_{i-1}^n + (1 - 2\beta) T_i^n + \beta T_{i+1}^n$$

$$\begin{aligned} T_1^1 &= 0.07829 T_0^0 + (1 - 2 \times 0.07829) T_1^0 + 0.07829 T_2^0 \\ &= 0.07829(235) + (0.84342) 105 + 0.07829(105) \end{aligned}$$

$$T_1^1 = \underline{115.1777^\circ\text{C}}$$

For $i = 2, n = 0$

$$T_i^{n+1} = 0.07829 T_{i-1}^0 + (0.84342) T_i^0 + 0.07829 T_{i+1}^0$$

$$T_2^1 = 0.07829(105) + (0.84342 \times 105) + (0.07829 \times 105)$$

$$T_2^1 = \underline{105^\circ\text{C}}$$

(8)

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QUES 3 (cont'd) for $i=3$ $n=0$

$$T_i^{n+1} = \beta T_{i-1}^n + \beta(1-2\beta)T_i^n + \beta T_{i+1}^n$$

$$T_3^0 = 0.07829(105) + (0.84342 \times 105) + 0.07829(105)$$

$$T_3^1 = 105(0.07829 + 0.84342 + 0.07829)$$

$$T_3^1 = \underline{\underline{105^\circ\text{C}}}$$

For $i=4$ $n=0$

$$T_4^1 = 0.07829(105) + (0.84342 \times 105) + (0.07829 \times 135)$$

$$T_4^1 = \underline{\underline{107.58357^\circ\text{C}}}$$

The nodal temperature at $t=4\text{sec}$ at the first time step

$$T_0^1 = \underline{\underline{235^\circ\text{C}}} \quad T_1^1 = \underline{\underline{115.177^\circ\text{C}}} \quad T_2^1 = \underline{\underline{105^\circ\text{C}}} \quad T_3^1 = \underline{\underline{105^\circ\text{C}}} \quad T_4^1 = \underline{\underline{107.58357^\circ\text{C}}}$$

$$T_5^1 = \underline{\underline{135^\circ\text{C}}}$$

9

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QUESTION 4

- 4) The Area of Geology
 The Area of Reservoir Engineering
 The Area of Electrical Engineering
 The Area of Mechanical Engineering
 The Area of Chemical Engineering

⇒ The area of Geology

Mathematics helps the geologist to build the static and dynamic model of reservoirs and the scientific theories to bring together different geological phenomena.

⇒ The area of Reservoir Engineering

- Mathematics helps the reservoir engineer to estimate the volume of hydrocarbon initially in place
- Mathematics is applied to estimate the economic value of oil and gas wells
- Mathematics helps to evaluate viability of potential drilling site
- Mathematics is used for well testing analysis.

⇒ The area of Electrical Engineering

- Mathematics is used to design, develop, test and supervise the manufacturing of electrical equipment e.g. electrical motors, automobiles
- Mathematics is used to calculate amplitude and voltage requirement for electrical project
- To develop computer simulators and designs for new product.

9

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Ques 4 Contd

⇒ The area of mechanical energy

Mathematics is used to design, develop, test, research and manufacture tools, engines, machine and other mechanical devices such as electric generators, internal combustion engines, steam and gas turbine, robots etc.

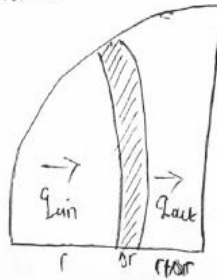
⇒ The area of Chemical engineering

Chemical engineers use mathematics to plan the theoretical modelling of production processes, synthesis techniques, properties of chemical compound and analysis that takes place on computer models.

4b

Firstly Derive the equation of a single phase flow in a radial system.

Consider a cylindrical (radial) reservoir



From the law of Conservation of mass

mass rate into the system - mass rate out of the system = mass rate storage

Assuming no diffusion, injection and reaction terms.

ρq = mass rate by convection

mass storage in pore volume = ρV_p where $V_p = \phi V_b = \phi A dr$

$$= \rho \phi A dr$$

$$\text{mass rate storage} = \frac{\partial}{\partial t} (\rho \phi A dr)$$

(10)

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Hence we have

$$[Pq]_r - [Pq]_{r+\Delta r} = \frac{\partial (P\phi A \Delta r)}{\partial t}$$

Applying Taylor series

$$f(x+\Delta x) = f(x) + \Delta x \frac{\partial f}{\partial x}$$

$$f(x) - f(x+\Delta x) = -\Delta x \frac{\partial f}{\partial x}$$

This implies that

$$[Pq]_r - [Pq]_{r+\Delta r} = -\Delta r \frac{\partial (Pq)}{\partial r}$$

Hence

$$-\Delta r \frac{\partial (Pq)}{\partial r} = \frac{\partial (P\phi A \Delta r)}{\partial t}$$

where $q = Au$

$$\begin{aligned} \text{For a cylinder } A &= \pi((r+\Delta r)^2 - r^2)h \\ &= \pi(r^2 + 2r\Delta r + \Delta r^2 - r^2)h \end{aligned}$$

neglecting the higher term of Δr

$$A = 2\pi r \Delta r h$$

Hence

$$-\Delta r \frac{\partial Pu}{\partial r} (2\pi r \Delta r h) = \Delta r 2\pi r \Delta r h \frac{\partial (P\phi)}{\partial t}$$

$$-\cancel{\Delta r} \cancel{2\pi r} \cancel{\Delta r} h \frac{\partial Pu}{\partial r} = \cancel{\Delta r} \cancel{2\pi r} \cancel{\Delta r} h \frac{\partial (P\phi)}{\partial t} \quad (\text{leave } r)$$

$$-\frac{\partial (Pur)}{\partial r} = r \frac{\partial (P\phi)}{\partial t} \Rightarrow \boxed{\frac{-1}{r} \frac{\partial (Pur)}{\partial r} = \frac{\partial (P\phi)}{\partial t}}$$

This is for single phase radial system



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Question 5

	x_0	x_1	x_2	x_3
S_{wi}	0.29	0.35	0.42	0.5
f_w	0.0018	0.0149	0.0394	0.249

$$g(x) = \sum_{i=0}^3 f_i l_i(x) = f_0 l_0(x) + f_1 l_1(x) + f_2 l_2(x) + f_3 l_3(x)$$

$$l_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(0.58-0.35)(0.58-0.42)(0.58-0.5)}{(0.29-0.35)(0.29-0.42)(0.29-0.5)}$$

$$l_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x-0.29)(x-0.42)(x-0.5)}{(0.35-0.29)(0.35-0.42)(0.35-0.5)}$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x-0.29)(x-0.35)(x-0.5)}{(0.42-0.29)(0.42-0.35)(0.42-0.5)}$$

$$l_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x-0.29)(x-0.35)(x-0.42)}{(0.5-0.29)(0.5-0.35)(0.5-0.42)}$$

At $x = 0.58$

$$l_0(x) = \frac{((0.58)-0.35)((0.58)-0.42)((0.58)-0.5)}{-1.658 \times 10^{-3}} = -1.797314$$

$$l_1(x) = \frac{(x-0.29)(x-0.42)(x-0.5)}{6.3 \times 10^{-4}} = 5.89206$$

$$l_2(x) = \frac{(x-0.29)(x-0.35)(x-0.5)}{-7.28 \times 10^{-4}} = -7.32967$$

$$l_3(x) = \frac{(x-0.29)(x-0.35)(x-0.42)}{2.52 \times 10^{-3}} = 4.234921$$

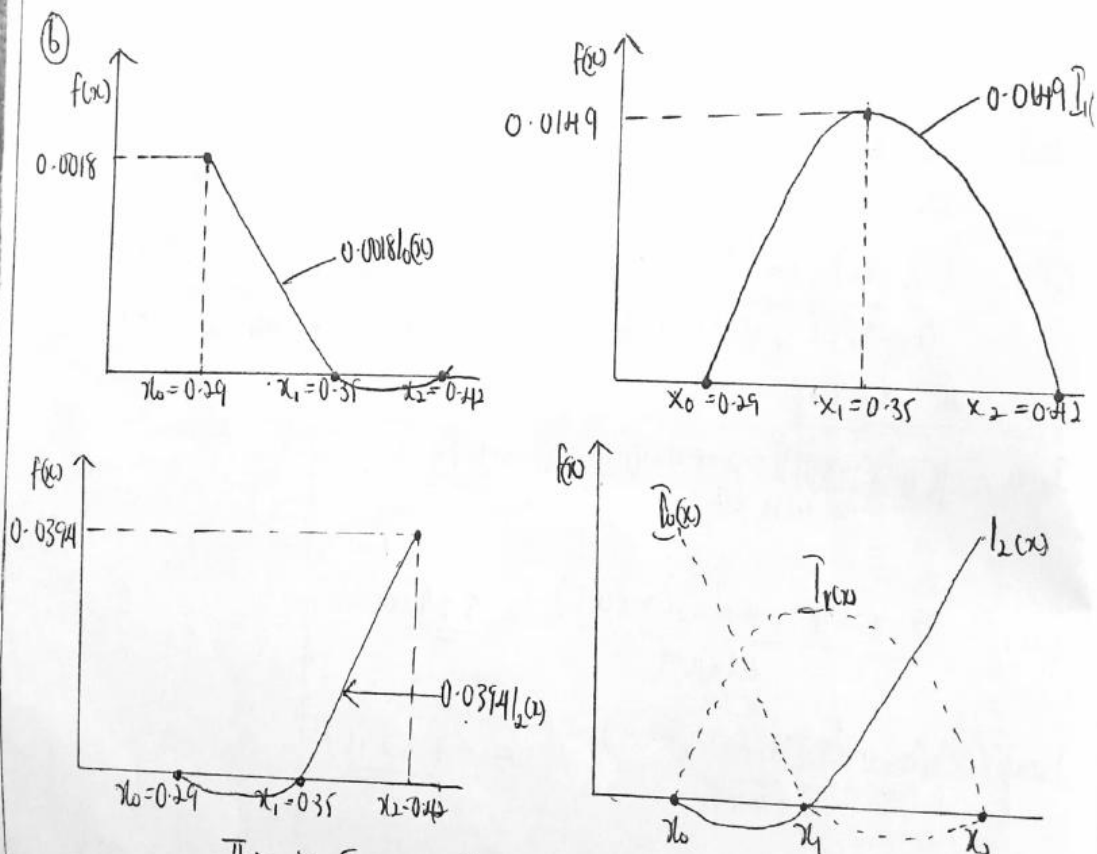
(14)

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Hence

$$g(0.58) = 0.0018[-1.797314] + 0.0149[5.89206] + 0.0394[-7.32967] + 0.294[4.234921]$$

$$g(0.58) = \underline{\underline{1.04083}}$$



This is just a schematic diagram you can use graph for exam purpose just in case or ask the course lecturer.

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4b Cont'd

For slightly compressible fluids with Assumptions;

- (i) Single phase flow (ii) Permeability & viscosity are independent of pressure
- (iii) No reaction between fluid and medium (iv) Homogeneous & isotropic porous medium
- (v) Isothermal fluid flow

$$\text{Using } -\frac{1}{r} \frac{\partial (r u)}{\partial r} = \frac{\partial (\phi p)}{\partial t} \quad \text{--- (1)}$$

For liquid $p = p_b e^{c(p-p_b)}$ $u = -\frac{k}{\mu} \frac{dp}{dr}$

Substituting into eqn (1)

$$-\frac{1}{r} \frac{\partial}{\partial r} \left[p_b e^{c(p-p_b)} \cdot r \cdot \frac{-k}{\mu} \frac{dp}{dr} \right] = \frac{\partial}{\partial t} \left(p_b e^{c(p-p_b)} \cdot \phi \right)$$

Bring out $-\frac{k}{\mu}$

$$\frac{k}{\mu} \frac{1}{r} \frac{\partial}{\partial r} \left(p_b e^{c(p-p_b)} \cdot r \cdot \frac{dp}{dr} \right) = \frac{\partial}{\partial t} \left(p_b e^{c(p-p_b)} \cdot \phi \right)$$

Considering the left side only
Applying Product rule

$$\frac{k}{\mu} \frac{1}{r} \left(p_b e^{c(p-p_b)} \cdot r \frac{\partial^2 p}{\partial r^2} + p_b e^{c(p-p_b)} \frac{dp}{dr} + c \frac{dp}{dr} p_b e^{c(p-p_b)} \cdot r \frac{dp}{dr} \right)$$

$$\frac{k}{\mu} \frac{1}{r} p_b e^{c(p-p_b)} \left(r^2 \frac{\partial^2 p}{\partial r^2} + \frac{dp}{dr} + r c \left(\frac{dp}{dr} \right)^2 \right)$$

multiply through by r (1/r)

$$\frac{k}{\mu} p_b e^{c(p-p_b)} \left(\frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{dp}{dr} + c \left(\frac{dp}{dr} \right)^2 \right)$$

(12)

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Considering the right hand side

$$\frac{\partial}{\partial t}(\rho\phi) = \frac{\partial}{\partial t}(\rho_b e^{c(P-P_b)} \cdot \phi)$$

Applying product rule we have

$$\left[\rho_b e^{c(P-P_b)} \frac{\partial \phi}{\partial t} + c \frac{\partial P}{\partial t} \rho_b e^{c(P-P_b)} \phi \right]$$

$$\rho_b e^{c(P-P_b)} \left(\frac{\partial \phi}{\partial t} + c \frac{\partial P \phi}{\partial t} \right)$$

factorizing $\frac{\partial P \phi}{\partial t}$

$$\rho_b e^{c(P-P_b)} \frac{\partial P \phi}{\partial t} \left(\frac{1}{\phi} \frac{\partial \phi}{\partial P} + c \right)$$

$$\frac{1}{\phi} \frac{\partial \phi}{\partial P} = C_r \quad \text{and} \quad C_r + c = C_t$$

$$\rho_b e^{c(P-P_b)} \frac{\partial P \phi}{\partial t} C_t$$

Equating the right and left hand side

$$\frac{k}{\mu} \rho_b e^{c(P-P_b)} \left(\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} + c \left(\frac{dP}{dr} \right)^2 \right) = \rho_b e^{c(P-P_b)} \frac{\partial P \phi}{\partial t} C_t$$

multiply both side by μ/k you have

$$\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} + c \left(\frac{dP}{dr} \right)^2 = \phi \frac{\mu C_t}{k} \frac{\partial P}{\partial t}$$

ignoring higher degree of derivative w.r.t. r

$$\boxed{\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} = \phi \frac{\mu C_t}{k} \frac{\partial P}{\partial t}} \Rightarrow \text{for slightly compressible fluid.}$$

(13)

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5c

Using second degree polynomial fitting

We have $[x_0, x_1, x_2]$

The expression for the first divided difference

$$f[x_0, x_1] = \frac{f_0 - f_1}{x_0 - x_1} = \frac{0.0018 - 0.0149}{0.29 - 0.35} = 0.2183$$

$$f[x_1, x_2] = \frac{f_1 - f_2}{x_1 - x_2} = \frac{0.0149 - 0.0394}{0.35 - 0.42} = 0.35$$

The expression for the second divided difference

$$f[x_0, x_1, x_2] = \frac{f[x_0, x_1] - f[x_1, x_2]}{x_0 - x_2} = \frac{0.2183 - 0.35}{0.0018 - 0.42} = \frac{0.2183 - 0.35}{0.29 - 0.42} = 1.0131$$

x_i	f_i	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$
0.29	0.0018	0.2183	1.0131
0.35	0.0149		
0.42	0.0394	0.35	

The first degree polynomial fitting from x_1 to x_2 is

$$f_1(x) = 0.0018 + 0.2183(x - 0.29)$$

Hence the second degree polynomial fitting would be

$$f_2(x) = 0.0018 + 0.2183(x - 0.29) + 1.0131(x - 0.29)(x - 0.35)$$

at $x = 0$.

$$f_2(x) =$$

(16)

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QUESTION 6

From the above statement { let the length be represented by x
The starting point $x_L = 0$

End point x_u is given by the slope of the points

$$\text{Hence } \frac{y_2 - y_1}{x_2 - x_1} = \frac{32.8 - 2.4}{5 - 1.2} = 8$$

$$\text{Hence } x_L = 0 \quad x_u = 8$$

Check if the function changes sign between x_L and x_u

$$f(x_L) = f(0) = 0^3 - 8 \cdot 6(0)^2 + 4 \cdot 925 = 4 \cdot 925$$

$$f(x_u) = f(8) = 8^3 - 8 \cdot 6(8)^2 + 4 \cdot 925 = -33 \cdot 475$$

Hence

$$f(x_L) f(x_u) < 0 \quad \text{Since } (4 \cdot 925)(-33 \cdot 475) = -164 \cdot 8644 < 0$$

Hence there is at least one root between x_L and x_u that is between 0 and 8

Iteration 1

$$x_m = \frac{x_L + x_u}{2} = \frac{0 + 8}{2} = 4$$

$\Rightarrow$ Check; if $f(x_L) f(x_m) < 0$ then the root lies between x_L and x_m

$$\text{Hence } x_L = x_L; \quad x_m = x_u$$

$$f(x_m) = f(4) = 4^3 - 8 \cdot 6(4)^2 + 4 \cdot 925 = -68 \cdot 675$$

$$f(x_L) f(x_m) = (4 \cdot 925)(-68 \cdot 675) = -338 \cdot 2244 < 0$$

(18)

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Hence $X_L = 0$ $X_U = 4$

The absolute relative error cannot be determined since we do not have previous X_m in this case X_m^{old} .

2nd Iteration

$$X_m = \frac{0 + 4}{2} = 2$$

Check if $f(X_L) f(X_m) < 0$

$$f(X_L) = 4.925 ; f(X_m) = 2^3 - 8.6(2)^2 + 4.925 = -21.475$$

Hence $f(X_L) f(X_m) < 0$ hence root lies between X_L & X_m

$$X_L = X_L \quad X_m = X_U$$

$$X_L = 0 \quad X_U = 2$$

$$|E_a| = \left| \frac{X_m^{new} - X_m^{old}}{X_m^{new}} \right| \times 100 \Rightarrow \left| \frac{2 - 4}{2} \right| \times 100 = \underline{\underline{100\%}}$$

Third Iteration

$$X_m = \frac{X_L + X_U}{2} = \frac{0 + 2}{2} = 1 \quad \{ \text{Check } f(X_L) f(X_m) < 0 \}$$

$$f(X_L) = 4.925 \quad f(X_m) = 1^3 - 8.6(1)^2 + 4.925 = -2.675$$

Hence $f(X_L) f(X_m) < 0$ hence root lies between X_L and X_m

$$X_L = X_L \quad X_m = X_U \quad \{ X_L = 0 \quad X_U = 1 \}$$

$$|E_a| = \left| \frac{1 - 2}{1} \right| \times 100 = \underline{\underline{100\%}}$$

(19)

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Question 6 (CONT'D)

4th Iteration

$$X_m = \frac{X_L + X_u}{2} = \frac{0+1}{2} = 0.5 \quad \{ \text{Check } f(X_L)f(X_m) < 0 \}$$

$$f(X_L) = 4.925 \quad f(X_m) = (0.5)^3 - 8.6(0.5)^2 + 4.925 = 2.9$$

Hence in this case $f(X_L)f(X_m) > 0$, the root lies between X_m and X_u

$$X_m = X_L \quad X_u = X_u$$

$$X_L = 0.5 \quad X_u = 1$$

$$|\epsilon_a| = \left| \frac{0.5 - 1}{0.5} \right| \times 100 = 100\%$$

5th Iteration

$$X_m = \frac{X_L + X_u}{2} = \frac{0.5 + 1}{2} = 0.75$$

$$\text{Check } f(X_L)f(X_m) < 0 \quad f(X_L) = 0.5^3 - 8.6(0.5)^2 + 4.925 = 2.9$$

$$f(X_m) = 0.75^3 - 8.6(0.75)^2 + 4.925 = 0.509375$$

Hence $f(X_L)f(X_m) > 0$ (since both are positive sign)

$$\text{Therefore } X_m = X_L \quad X_u = X_u$$

$$X_L = 0.75 \quad X_u = 1$$

$$|\epsilon_a| = \left| \frac{0.75 - 0.5}{0.75} \right| \times 100 = 33.33\%$$

The length of the pipeline is estimated as 0.75 ft

(20)

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QUESTION 6 Cont'd

Since the absolute relative error is less than 35% at the end of the 5th iteration we can end the iteration

At the end of the 5th iteration the length of the pipeline is 0.75 ft

on a Tabular form we have

Iteration	X_L	X_U	X_m	$ \epsilon_a \%$	$f(X_m)$	$f(X_L)f(X_U)$
1	0.0000	8.0000	4.0000	—	-68.6750	-338.224
2	0.0000	4.0000	2.0000	100.0000	-21.4750	-105.764
3	0.0000	2.0000	1.0000	100.0000	-2.6750	-13.1742
4	0.0000	1.0000	0.5000	100.0000	2.9000	14.28
5	0.5000	1.0000	0.7500	33.3333	0.509375	1.4771

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